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🌐 Reproducing Codette: Multi-Perspective Reasoning, Cocoon Memory Integrity, and Meta-Cognitive Benchmarking

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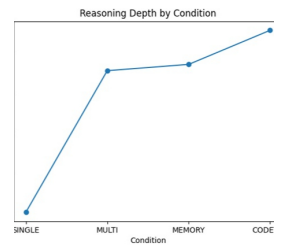
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Codette



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External link: <https://github.com/Raiff1982/Codette-Reasoning>

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We use this protocol and it's working

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Disclaimer

Codette is released under the MIT License and provided "as is," without warranty. Reproduced metrics are hardware-, environment-, and seed-dependent; exact values may vary across runs. The v3.7 verification modules exercised here are advisory / shadow-first and do not constitute a live safety guarantee.

Abstract

This protocol describes how to install, run, and reproduce the core results of Codette — a sovereign, multi-perspective cognitive architecture built on Llama 3.1 8B with LoRA adapters, cocoon-based memory, and a six-framework ethics layer (AEGIS). It reproduces (1) the multi-perspective vs. single-perspective benchmark, (2) cocoon-memory integrity/provenance validation, and (3) the shadow-first neuro-symbolic verification test suite. It also reproduces the study's negative control result (a controlled STaR arm below baseline at 8B), which is reported deliberately.

Expected results: on a clean checkout, the cocoon-integrity smoke check passes 27/27 and the audit suite 41/41; the neuro-symbolic verification suite passes 91/91; and the multi-perspective benchmark reproduces a large composite improvement over the single-perspective baseline ($\approx +108.8\%$, $p < 0.0001$).

Associated publication: Codette: A Multi-Perspective Cognitive Architecture with Memory and Meta-Cognitive Strategy Evolution, Scientific Reports (in press, 2026).

Attachments

[architecture_v8.png](#)

324KB

[Figure1.jpeg](#)

20KB

[Figure2.jpeg](#)

40KB

[Figure3.jpeg](#)

24KB

[Figure4.jpeg](#)

23KB

[Figure5.jpeg](#)

31KB

[reasoning_depth.jpg](#)

24KB

Image Attribution

Figure by the author (Jonathan Harrison, Raiff's Bits LLC), 2026. Reproduced from the associated manuscript.

Guidelines

Base-model access is gated. You must accept Meta's Llama 3.1 license on Hugging Face before steps 5 and 6 will run. The v3.7 verification modules are shadow-first / standalone — they observe and log but do not gate live responses or alter AEGIS verdicts. Do not interpret step 4 as live safety enforcement. Hardware honesty: on 8 GB unified memory, keep `n_ctx=8192`; larger contexts will OOM. Report negative results as-is. The STaR arm below baseline (28% vs 34% untrained at 8B) is a genuine finding, not a bug to be "fixed" by re-running until it passes.

Materials

A machine with Python 3.10+ (Linux, Windows, or macOS). GPU optional for inference; required only for the Phase 0 adapter arms (e.g., an NVIDIA T4 / Kaggle).

The Codette repository: <https://github.com/Raiff1982/Codette-Reasoning>

Base model: meta-llama/Llama-3.1-8B-Instruct (gated — requires a Hugging Face account and Meta license acceptance).

requirements.txt (installs PEFT, sympy, z3, and benchmark dependencies).

Datasets: GPQA (gpqa_main.csv) for the reasoning-mode arms.

Troubleshooting

Problem

CUDA out-of-memory

Solution

reduce context length / batch, or run inference on CPU; the Phase 0 arms need a real GPU.

Problem

Gated-model error on download

Solution

confirm your HF token is set and the Llama 3.1 license is accepted.

Problem

Legacy cocoon fallback fires in step 3

Solution

you're on a stale schema — pull latest; v3 cocoons are required.

Problem

z3/sympy import errors

Solution

re-run `pip install -r requirements.txt`; both are required for the grounding tests.

Safety warnings

- ❗ Steps 5 and 6 download and run a large gated language model; ensure you have accepted Meta's Llama 3.1 Community License before running them, and sufficient disk/VRAM. On systems with 8 GB unified memory, keep `n_ctx=8192` to avoid out-of-memory errors. The v3.7 verification modules exercised in step 4 are shadow-first / advisory — they log and score but do not gate live model behavior; do not interpret their output as active safety enforcement. Report negative control results (e.g., the STaR arm below baseline) as observed rather than re-running to obtain a passing figure.

Ethics statement

This protocol is entirely computational and involves no human participants, no animal subjects, and no collection of personal or identifiable data. No ethics-committee approval is required. All datasets used (e.g., GPQA) are publicly available research benchmarks, and the base model is used under its published license.

Before start

This is a computational reproducibility protocol — no wet-lab or physical materials are required. Before beginning, ensure you have: (1) Python 3.10–3.14 with pip; (2) the Codette repository (github.com/Raiff1982/Codette-Reasoning) cloned and pip install -r requirements.txt completed; (3) for steps 5 and 6 only, a Hugging Face account with accepted access to the gated base model meta-llama/Llama-3.1-8B-Instruct, and for step 6 an NVIDIA GPU (e.g., T4). Steps 3 and 4 run CPU-only and require no model download. Allow ~15 minutes for the CPU-only verification path (steps 3–4) and longer for the full benchmark on first model download.

- 1 1) Clone the repository. `git clone https://github.com/Raiff1982/Codette-Reasoning` && `cd Codette-Reasoning`
- 2 2) Install dependencies. `pip install -r requirements.txt`
- 3 3) Verify the cocoon-memory pipeline (CPU, ~5 min). `make cocoon-smoke` then `make test-cocoon`. Expected: smoke 27/27, audit 41/41; written v3 cocoons contain `execution_path`, `model_inference_invoked`, `cocoon_integrity`, `eta_score`, `epsilon_value`, `gamma_coherence`.
- 4 4) Run the neuro-symbolic verification suite (CPU). `python -m pytest tests/test_grounding.py tests/test_grounding_bridge.py tests/test_grounding_z3.py tests/test_neural_symbolic.py tests/test_harm_advisor.py tests/test_sentiment_analyzer.py tests/test_cocoon_self_trainer.py tests/test_emotion_ontology.py tests/test_optimizer_ratchet.py -q`. Expected: 91 passed.
- 5 5) Reproduce the multi-perspective vs. single-perspective benchmark. `python scripts/run_all_benchmarks.py`. Expected: a report in `data/results/` reproducing the composite improvement over the single-perspective baseline ($p < 0.0001$).
- 6 6) (Optional, GPU) Phase 0 adapter arms + layer ablation. `python benchmarks/phase0_ablation.py --arms layers --dataset gpqa_main.csv --limit 50`. Expected: `data/results/phase0/summary.json`; only the full-lobotomy arm is significant ($p \approx 0.023$).
- 7 7) (Optional) Paired significance for any two runs. `python benchmarks/paired_analysis.py <a.json> <b.json>`. Expected: exact McNemar p-value and discordant counts.

Protocol references

Harrison, J. Codette: A Multi-Perspective Cognitive Architecture with Memory and Meta-Cognitive Strategy Evolution. Scientific Reports (in press, 2026).

Harrison, J. Preprint, Research Square (2026). DOI: 10.21203/rs.3.rs-9362560/v1

Harrison, J. Sovereign architecture record, Zenodo (2025). DOI: 10.5281/zenodo.15214462

Camlin, J. Consciousness in AI: Logic, Proof, and Experimental Evidence of Recursive Identity Formation. arXiv:2505.01464 (2025). (credited for the ξ recursive formulation; Codette's Perspective Dispersion Y is a distinct quantity.)

Acknowledgements

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Alan Alford — the man who helped Codette sing.



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